SMARTBRIDGE
Let's Bridge the Gap

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	xxxxxx
Project Title	xxxxxx
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Analyze Formula 1 results and race/lap data in Tableau to compare driver and constructor performance, tyre usage, podiums and reliability.
Scope	Five supplied sheets; seasons 2010–2026 as represented; 6,464 race-result rows. Scope is limited to fields available in the workbook.
Race performance information is spread across result, standings, qualifying, circuit and lap-level tables, making comparison difficult without a consistent analytical view.	
Description	Users need a reliable, interactive way to answer strategy, broadcast and technical-reliability questions without manually scanning spreadsheets.
Impact	Supports evidence-led discussion of performance trends and reliability; results remain descriptive and should not be presented as causal proof.
Proposed Solution	
Approach	Inspect workbook, validate grain/types, relate tables, create calculated fields, build worksheets and dashboards, validate totals, then assemble a Tableau Story.
Key Features	Season/team/driver filters; points leaderboard; constructor trends; tyre compound and tyre-age analysis; podiums; DNF/reason analysis; circuit and weather comparisons.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard laptop/desktop capable of running Tableau Desktop

Memory	RAM specifications	8 GB RAM recommended; 16 GB helpful for smoother workbook use
Storage	Disk space for data, models, and logs	At least 2 GB free for source workbook, Tableau workbook and exports
Software		
Frameworks	Python frameworks	Tableau Desktop / Tableau Public; Snowflake optional only if deployed
Libraries	Additional libraries	Excel workbook; optional Python pandas/openpyxl for validation
Development Environment	IDE, version control	Tableau Desktop, Excel-compatible environment, project ZIP for documentation
F1 Racing Project Dataset(1).xlsx; five sheets; 6,464 race_results rows plus supporting sheets		
Data	Source, size, format	F1 Racing Project Dataset(1).xlsx; five sheets; 6,464 race_results rows plus supporting sheets

Project-specific Documentation

Dataset workbook: F1 Racing Project Dataset(1).xlsx.
Sheets and data rows (excluding headers): race_results 6,464; circuits 29; championship_standings 311; lap_intelligence 3,814; qualifying_results 3,130.
Observed race_results coverage: seasons 2010–2026; 334 distinct race_id values; 19 distinct driver labels; 10 distinct team labels; 21 distinct country labels.
Available analysis fields include race date/season/round, circuit type and location, driver/team, finish/grid position, points, DNF and DNF reason, weather/wet-race flag, safety-car/VSC/red-flag data, pit stops, tyre strategy/starting compound, overtakes and laps led.
Lap-level fields include lap/sector times, tyre compound and tyre age, DRS, speed trap and pit-lap flag.
Qualifying fields include qualifying position, Q1/Q2/Q3 times, gap to pole and top speed.
Validation note: workbook is a multi-sheet analytical dataset, not necessarily official complete historical timing. Verify values and data types before presenting findings. Do not infer driver nationality: no nationality field is present.
Use COUNTD([race_id]) for race-event counts where appropriate; race_results contains multiple driver-result rows per event. Distinguish race-result row counts from unique Grand Prix counts.

- Recommended analysis: driver points; constructor season trajectory; tyre compound usage; tyre age versus lap time; podium distribution; country/circuit comparisons; DNF by team/season/reason; wet-race/circuit-type reliability; grid-to-finish position gain; pit-stop and safety-car context.
- Limitations: no driver nationality field; DNF reasons are sparse; dataset's historical coverage and synthetic-looking labels should be validated against an authoritative source before external claims.